

# ECCA Roadside Litter Study

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Mark Bennett  
Addison Cartmell  
Maddi Gao  
Khloud Zamzani

### Introduction:

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### About the Project

This project was developed in collaboration with **East County Citizens' Alliance (ECCA)**, an organization that focuses on environmental issues and roadside litter in the Portland area.

## **ECCA Roadside Litter Study**

### **Introduction:**

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#### **Project Goal**

The goal of the project is to develop a practical system for studying roadside litter and understanding:

- What types of litter are most common
- How much litter accumulates
- How litter changes over time

#### **Focus Areas**

- Plastic pollution
- Alcohol-related litter
- Unsecured-load debris
- Other roadside waste affecting the environment and public safety

ECCA Roadside Litter Study

# Sampling Design

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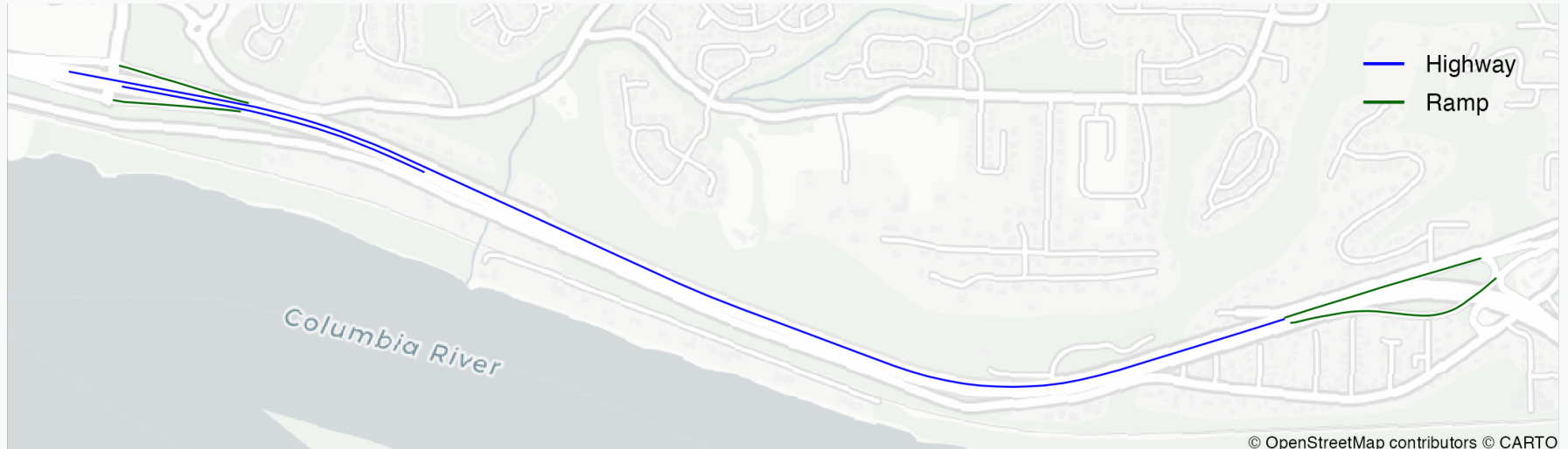
# Approach

**Determine areas (strata) to use to group all the highways and ramps**

**Assume trash in a given area is relatively similar anywhere in the area, but possibly different from other areas**

**Pick some number of segments from each area to use in the sample**

# West Side Areas



- 2 highway stretches (north/westbound, south/eastbound)
- 2 pairs of ramps

# East Side Areas



- 3 highway stretches
- 1 pair of ramps

# Sampling Execution

## Idea

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Do sampling entirely in software using geospatial data from WSDOT

We use data from the **WSDOT Geospatial Open Data Portal**:

- Highway data: **WSDOT - State Route Linear Referencing System (LRS) Current**
- Ramp data: **WSDOT - State Route Ramp (1:24K) Current**

# Highway Sampling

1. Load geospatial data for both sides of Highway 14

2. Filter these into the 5 areas using the milepost boundaries for what ECCA cleans

3. Divide each area into 600 foot segments

4. Randomly select one of these segments in each area

# Ramp Sampling

## Idea

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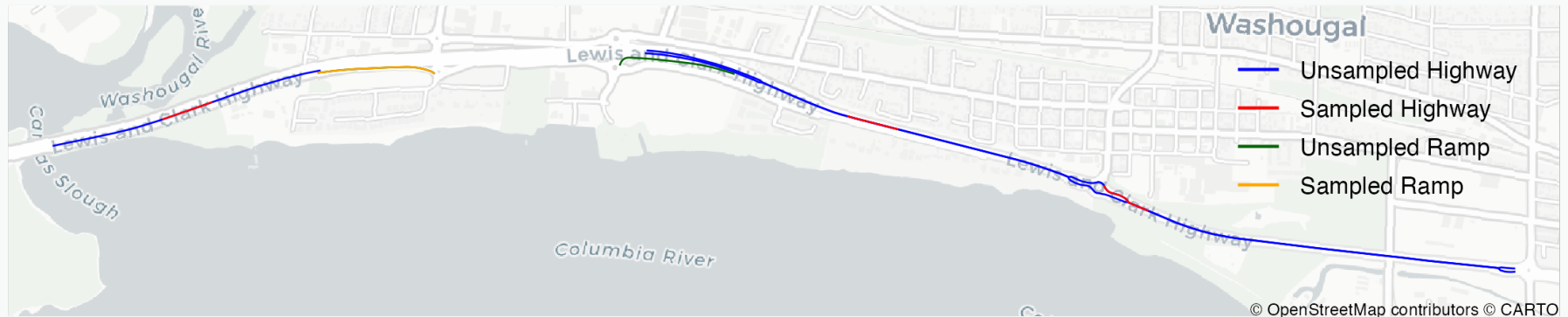
Pick one ramp from each pair, but make sure at least one on-ramp and one off-ramp are selected

# Sampling Results: West Side



- 2 highway samples
- 2 ramp samples

# Sampling Results: East Side



- 3 highway samples
- 1 ramp sample

## ECCA Roadside Litter Study

# Categorization Design

MATERIAL (MUTUALLY EXCLUSIVE — PRIMARY MATERIAL BY VOLUME)

Paper / Cardboard

Plastic

Glass

Metal

Wood

Textile

Other

FUNCTIONAL CATEGORIES (SELECT ALL THAT APPLY)

Food Container

Toxic Fluid Container

Automotive Debris

Construction / Demolition

Lawn & Garden

Marina-Related

Attire

Furniture

Lumber Tag

Table 72. Detailed litter composition table for Interstate roadways.

| Material                                                        | Est. % | +/-  | Est. Pounds per Mile | Pounds per Mile +/- | Est. % | +/-  | Est. Pounds per Mile | Pounds per Mile +/- | Est. Pounds per Mile | Pounds per Mile +/- |
|-----------------------------------------------------------------|--------|------|----------------------|---------------------|--------|------|----------------------|---------------------|----------------------|---------------------|
| Paper                                                           | 4.2%   | 0.6% | 59.1                 | 9.4                 | 15.4%  | 1.2% | 4,374                | 478                 | 191.5                | 8,367               |
| Cardboard                                                       | 0.2%   | 0.0% | 24.2                 | 3.2                 | 0.2%   | 0.0% | 881                  | 29                  | 51.2                 | 1,111               |
| Fast-food Paper Bags                                            | 0.0%   | 0.0% | 1.1                  | 0.2                 | 0.0%   | 0.0% | 79                   | 18                  | 1.9                  | 46                  |
| Fast Food Paper Bags                                            | 0.0%   | 0.0% | 8.4                  | 0.6                 | 0.0%   | 0.0% | 289                  | 60                  | 2.0                  | 46                  |
| Fast-food Paper Cups                                            | 0.4%   | 0.1% | 1.1                  | 0.2                 | 0.4%   | 0.1% | 164                  | 50                  | 4.0                  | 109                 |
| Cups                                                            | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 1.1                  | 42                  |
| One-Trial To-Go/Fast Food Service Items                         | 0.1%   | 0.0% | 1.5                  | 0.6                 | 0.1%   | 0.1% | 393                  | 30                  | 3.8                  | 138                 |
| Paper Napkins and Paper Plates                                  | 0.1%   | 0.0% | 1.6                  | 0.6                 | 0.1%   | 0.1% | 218                  | 26                  | 8.3                  | 1,444               |
| Paper Menus and Food Cartons, Alcoholic                         | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.0%   | 0.0% | 3                    | 3                   | 0.2                  | 5                   |
| Paper Beverages and Food Cartons, Non-Alcoholic                 | 0.0%   | 0.0% | 0.2                  | 0.1                 | 0.0%   | 0.0% | 23                   | 6                   | 0.5                  | 207                 |
| Paper Non-Beverage Paper/Fast Food Packaging                    | 0.1%   | 0.1% | 3.1                  | 0.8                 | 0.0%   | 0.0% | 259                  | 76                  | 2.9                  | 193                 |
| Non-food Paper Packaging                                        | 0.1%   | 0.1% | 3.1                  | 1.0                 | 0.1%   | 0.1% | 281                  | 76                  | 7.6                  | 446                 |
| Newspapers, Magazines, Leaf Mail, and Office Paper              | 0.2%   | 0.0% | 7.7                  | 2.7                 | 0.2%   | 0.0% | 1,498                | 267                 | 18.4                 | 1,110               |
| Household                                                       | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.1%   | 0.1% | 36                   | 22                  | 17.1                 | 739                 |
| Other Paper                                                     | 0.2%   | 0.0% | 8.0                  | 1.0                 | 1.2%   | 0.4% | 1,710                | 337                 | 16.1                 | 2,739               |
| Plastics                                                        | 13.1%  | 1.0% | 151.0                | 18.2                | 38.6%  | 2.9% | 14,844               | 1,519               | 118.1                | 18,770              |
| Plastic Bottles and Lids, Beer                                  | 0.0%   | 0.0% | 3.9                  | 0.6                 | 0.0%   | 0.0% | 7                    | 3                   | 0.9                  | 0                   |
| Plastic Bottles and Lids, Soda                                  | 0.0%   | 0.1% | 2.4                  | 0.7                 | 0.0%   | 0.1% | 195                  | 48                  | 2.9                  | 92                  |
| Plastic Bottles and Lids, Single-serve Wine and Liquor          | 0.1%   | 0.0% | 1.1                  | 0.4                 | 0.0%   | 0.1% | 92                   | 36                  | 0.8                  | 21                  |
| Plastic Bottles and Lids, Other Wine and Liquor                 | 0.0%   | 0.0% | 0.1                  | 0.2                 | 0.0%   | 0.0% | 8                    | 5                   | 1.1                  | 7                   |
| Plastic Bottles and Lids, Sports and Health Drinks              | 0.0%   | 0.0% | 0.4                  | 0.2                 | 0.0%   | 0.0% | 34                   | 12                  | 2.6                  | 14                  |
| Plastic Bottles and Lids, Juice                                 | 0.2%   | 0.1% | 2.6                  | 1.0                 | 0.2%   | 0.1% | 66                   | 28                  | 0                    | 0                   |
| Plastic Bottles and Lids, Tea and Coffee                        | 0.0%   | 0.0% | 0.3                  | 0.2                 | 0.0%   | 0.0% | 14                   | 7                   | 1.2                  | 13                  |
| Plastic Bottles and Lids, Still-water                           | 0.4%   | 0.1% | 4.7                  | 0.9                 | 1.2%   | 0.2% | 479                  | 69                  | 8.8                  | 414                 |
| Plastic Bottles and Lids, Other water                           | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.0%   | 0.0% | 3                    | 2                   | 1.9                  | 36                  |
| Plastic Bottles and Lids, Unknown                               | 0.0%   | 0.0% | 4.2                  | 2.9                 | 0.1%   | 0.0% | 27                   | 18                  | 1.1                  | 18                  |
| Plastic Cans and Closures                                       | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.0%   | 0.1% | 282                  | 18                  | 6.1                  | 170                 |
| Plastic Bottle Caps and Tabs                                    | 0.0%   | 0.0% | 0.2                  | 1.6                 | 0.0%   | 0.0% | 346                  | 40                  | 2.3                  | 490                 |
| Plastic Beverage Rings                                          | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 3                    | 3                   | 0.1                  | 7                   |
| Fast-food Plastic Caps                                          | 0.2%   | 0.1% | 3.5                  | 1.1                 | 0.0%   | 0.1% | 218                  | 42                  | 3.5                  | 115                 |
| Plastic Caps                                                    | 0.1%   | 0.0% | 0.7                  | 0.3                 | 0.1%   | 0.1% | 93                   | 73                  | 0.8                  | 102                 |
| Plastic Lids                                                    | 0.1%   | 0.1% | 1.7                  | 0.6                 | 0.1%   | 0.1% | 284                  | 79                  | 0.1                  | 3                   |
| Plastic Utensils                                                | 0.0%   | 0.0% | 0.4                  | 0.4                 | 0.1%   | 0.1% | 98                   | 47                  | 0.2                  | 26                  |
| Plastic, Resin and Single-use Containers                        | 0.0%   | 0.0% | 0.9                  | 0.9                 | 0.0%   | 0.0% | 0                    | 0                   | 1.5                  | 96                  |
| Clay/Clayware                                                   | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.0%   | 0.0% | 0                    | 0                   | 0.9                  | 0                   |
| Plastic Trash Bags                                              | 0.0%   | 0.1% | 2.3                  | 1.0                 | 0.2%   | 0.0% | 95                   | 38                  | 2.4                  | 103                 |
| Grocery, Retail and Shipping Bags                               | 0.1%   | 0.0% | 0.1                  | 0.4                 | 0.1%   | 0.1% | 10                   | 12                  | 1.3                  | 134                 |
| Other Plastic Film                                              | 1.1%   | 0.3% | 18.7                 | 10.5                | 1.2%   | 1.2% | 1,869                | 461                 | 9.0                  | 2,411               |
| Food Wrappers and Trash Bags                                    | 0.1%   | 0.1% | 8.9                  | 1.2                 | 0.0%   | 0.0% | 1,191                | 199                 | 7.9                  | 1,311               |
| Food and Drink Packaging                                        | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.0%   | 0.0% | 16                   | 7                   | 0.5                  | 82                  |
| Food Cans, Resin, and Closures                                  | 0.0%   | 0.0% | 9.8                  | 0.3                 | 0.0%   | 0.0% | 187                  | 20                  | 0.3                  | 82                  |
| Other Resin                                                     | 0.0%   | 0.2% | 9.8                  | 2.9                 | 4.2%   | 0.7% | 1,611                | 268                 | 8.7                  | 1,818               |
| Non-Beverage Plastic/Food Packaging                             | 0.2%   | 0.1% | 2.3                  | 0.7                 | 0.1%   | 0.1% | 387                  | 26                  | 2.1                  | 120                 |
| Non-food Plastic Packaging                                      | 0.2%   | 0.1% | 3.7                  | 1.3                 | 0.0%   | 0.0% | 125                  | 48                  | 12.4                 | 2,345               |
| Other Plastic                                                   | 0.1%   | 1.1% | 73.4                 | 125                 | 12.4%  | 1.4% | 4,778                | 561                 | 46.3                 | 8,479               |
| Glass                                                           | 1.3%   | 0.4% | 16.1                 | 4.8                 | 1.9%   | 0.8% | 511                  | 123                 | 38.8                 | 1,597               |
| Glass Bottles and Lids, Beer                                    | 0.1%   | 0.0% | 8.8                  | 4.1                 | 0.2%   | 0.1% | 15                   | 8                   | 18.1                 | 161                 |
| Glass Bottles and Lids, Soda                                    | 0.0%   | 0.0% | 0.9                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.8                  | 0                   |
| Glass Bottles and Lids, Single-serve Wine and Liquor            | 0.0%   | 0.0% | 0.1                  | 0.0                 | 0.0%   | 0.0% | 0                    | 2                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Other Wine and Liquor                   | 0.0%   | 0.0% | 0.1                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Sports and Health Drinks                | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Juice                                   | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Tea and Coffee                          | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Still-water                             | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Unknown                                 | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Glass Bottles and Lids, Other Water                             | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Other Glass/Food Packaging                                      | 0.1%   | 0.1% | 1.1                  | 0.8                 | 0.1%   | 0.1% | 30                   | 19                  | 2.1                  | 86                  |
| Other Glass/Non-Food Packaging                                  | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Other Glass or Ceramic                                          | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Other Glass                                                     | 0.1%   | 0.1% | 1.1                  | 0.7                 | 0.1%   | 0.2% | 213                  | 19                  | 4.1                  | 1,110               |
| Other Glass                                                     | 0.1%   | 0.1% | 2.0                  | 1.9                 | 0.1%   | 0.1% | 111                  | 12                  | 0.1                  | 8                   |
| Metal                                                           | 6.8%   | 1.0% | 83.5                 | 22.8                | 5.1%   | 0.6% | 1,977                | 238                 | 18.1                 | 3,438               |
| Metal Bottles, Lids and Cans, Beer                              | 0.4%   | 0.1% | 4.7                  | 1.1                 | 1.0%   | 0.1% | 491                  | 118                 | 8.7                  | 184                 |
| Metal Bottles, Lids and Cans, Soda                              | 0.2%   | 0.0% | 1.9                  | 0.1                 | 0.0%   | 0.0% | 189                  | 10                  | 2.0                  | 96                  |
| Metal Bottles, Lids and Cans, Sports and Health Drinks          | 0.1%   | 0.1% | 2.1                  | 1.1                 | 0.1%   | 0.1% | 161                  | 27                  | 1.2                  | 110                 |
| Metal Bottles, Lids and Cans, Juice                             | 0.0%   | 0.0% | 0.1                  | 0.0                 | 0.0%   | 0.0% | 8                    | 2                   | 0.1                  | 5                   |
| Metal Bottles, Lids and Cans, Tea and Coffee                    | 0.0%   | 0.0% | 0.1                  | 0.2                 | 0.1%   | 0.0% | 46                   | 10                  | 1.9                  | 25                  |
| Metal Bottles, Lids and Cans, Still-water                       | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Metal Bottles, Lids and Cans, Other Water                       | 0.1%   | 0.0% | 1.2                  | 0.9                 | 0.0%   | 0.0% | 13                   | 9                   | 0.3                  | 12                  |
| Metal Bottles, Lids and Cans, Unknown                           | 0.0%   | 0.0% | 0.2                  | 0.1                 | 0.1%   | 0.0% | 29                   | 11                  | 0.2                  | 10                  |
| Non-Beverage Metal/Food Packaging                               | 0.1%   | 0.1% | 2.9                  | 0.8                 | 0.0%   | 0.0% | 128                  | 18                  | 2.8                  | 810                 |
| Metal Non-Food Packaging                                        | 0.1%   | 0.0% | 0.4                  | 0.4                 | 0.1%   | 0.0% | 30                   | 18                  | 1.4                  | 443                 |
| Scrap Cans and Lids                                             | 0.1%   | 0.1% | 2.1                  | 1.1                 | 0.1%   | 0.1% | 128                  | 19                  | 1.1                  | 187                 |
| Other Metal                                                     | 5.1%   | 1.0% | 66.2                 | 22.6                | 1.4%   | 0.4% | 543                  | 146                 | 14.4                 | 1,124               |
| Smoking                                                         | 1.4%   | 0.3% | 177.6                | 4.8                 | 5.0%   | 0.6% | 1,593                | 269                 | 19.0                 | 13,444              |
| Cigarette Butts                                                 | 0.0%   | 0.1% | 4.2                  | 1.7                 | 1.4%   | 0.0% | 1,139                | 109                 | 11.8                 | 12,817              |
| Other Cigarettes, Tobacco, and Cigarette Products and Packaging | 0.0%   | 0.0% | 0.4                  | 1.2                 | 1.4%   | 0.0% | 146                  | 19                  | 1.4                  | 199                 |
| Cigarettes and Tobacco                                          | 0.0%   | 0.0% | 7.8                  | 5.9                 | 0.1%   | 0.1% | 64                   | 17                  | 0.5                  | 118                 |
| Food Waste                                                      | 0.2%   | 0.0% | 102.8                | 38.2                | 1.1%   | 0.1% | 426                  | 127                 | 27.1                 | 3,915               |
| Human or Pet Waste                                              | 0.1%   | 0.1% | 1.7                  | 1.2                 | 0.1%   | 0.0% | 10                   | 14                  | 0.0                  | 0                   |
| Other Organics                                                  | 0.1%   | 4.0% | 184.7                | 50.1                | 0.0%   | 0.0% | 739                  | 19                  | 18.8                 | 2,114               |
| Construction and Demolition Debris                              | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| CSL Waste                                                       | 0.0%   | 1.6% | 118.7                | 18.8                | 5.1%   | 0.6% | 1,961                | 240                 | 168.7                | 2,819               |
| Construction and Demolition Debris                              | 0.1%   | 1.4% | 114.7                | 18.4                | 1.1%   | 0.4% | 1,161                | 149                 | 16.1                 | 2,149               |
| Hazardous Materials                                             | 0.7%   | 0.4% | 8.0                  | 3.9                 | 8.8%   | 0.1% | 224                  | 135                 | 2.6                  | 227                 |
| Chemicals, Pests and Other Insecticide Materials                | 0.1%   | 0.4% | 6.7                  | 4.0                 | 0.1%   | 0.0% | 22                   | 10                  | 0.0                  | 0                   |
| Medical Waste                                                   | 0.1%   | 0.1% | 0.8                  | 0.7                 | 0.2%   | 0.1% | 131                  | 198                 | 0.8                  | 146                 |
| PEE Waste                                                       | 0.0%   | 0.0% | 0.1                  | 0.1                 | 0.2%   | 0.0% | 37                   | 6                   | 0.7                  | 75                  |
| Explosives (Shave gel, Firecrackers or semtex-like)             | 0.0%   | 0.0% | 0.9                  | 0.0                 | 0.0%   | 0.0% | 5                    | 7                   | 1.2                  | 1                   |
| Other Materials                                                 | 55.1%  | 1.1% | 683.1                | 61.2                | 31.8%  | 4.5% | 12,216               | 1,307               | 388.4                | 24,101              |
| Fishing and Maritime Items                                      | 0.2%   | 0.1% | 2.0                  | 1.3                 | 0.1%   | 0.1% | 53                   | 13                  | 0.9                  | 9                   |
| Tires                                                           | 0.1%   | 0.1% | 44.1                 | 30.7                | 0.0%   | 0.0% | 6                    | 6                   | 0.1                  | 11.5                |
| Auto Rubber Products (Stumps, tire shreds or treads)            | 22.7%  | 4.4% | 271.9                | 12.6                | 18.3%  | 1.3% | 7,045                | 1,167               | 10.4                 | 11,798              |
| Vehicle Debris                                                  | 0.1%   | 0.0% | 288.1                | 17.9                | 0.1%   | 0.1% | 1,197                | 147                 | 18.1                 | 1,816               |
| Batteries, Electronics, and Small Appliances                    | 0.1%   | 0.0% | 1.1                  | 0.5                 | 0.1%   | 0.0% | 21                   | 18                  | 1.5                  | 15                  |
| Large Appliances                                                | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 0.0                  | 0                   |
| Household/Commercial Items/Office Items                         | 1.0%   | 0.0% | 12.4                 | 5.5                 | 0.0%   | 0.0% | 185                  | 19                  | 25.5                 | 424                 |
| Toilets, Clothing and Shoes                                     | 0.0%   | 0.0% | 43.1                 | 20.2                | 0.0%   | 0.0% | 145                  | 14                  | 24.8                 | 1,144               |
| Toiletries and Personal Products                                | 0.2%   | 0.1% | 2.8                  | 0.8                 | 1.1%   | 0.0% | 461                  | 102                 | 0.5                  | 173                 |
| Batteries                                                       | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 6                    | 3                   | 0.0                  | 2                   |
| Toys, Sports, and Recreational Items                            | 0.1%   | 0.0% | 0.7                  | 0.4                 | 0.1%   | 0.1% | 40                   | 10                  | 2.7                  | 55                  |
| Furniture, Carpets, and Bedding Items                           | 0.0%   | 0.0% | 0.0                  | 0.0                 | 0.0%   | 0.0% | 0                    | 0                   | 2.8                  | 10                  |
| Whole Bags of Mixed Trash                                       | 0.3%   | 0.3% | 3.8                  | 3.0                 | 0.0%   | 0.0% | 6                    | 5                   | 18.1                 | 12                  |
| Miscellaneous Materials                                         | 0.1%   | 1.1% | 11.0                 | 10.8                | 0.1%   | 0.1% | 116                  | 71                  | 1.7                  | 1,211               |
| Estimated Total (per mile)                                      | 100%   |      | 1,127.0              | 129.6               | 100%   |      | 18,518               | 1,578.8             | 173.68               | 21,519              |

Figure 105. Sample sorting, full sample (left), sub sample (right).



# ECCA's Perspective

**ECCA has its own ideas about what data to collect**

**Weight doesn't give us the picture we're looking to capture**

**The data collection process can't be a terrible headache**

# Our Solutions

**Use an applet for item-by-item data entry**

**Weigh total trash, alongside selected trash categories**

# Conclusion

The categories are highly adapted to ECCA's concerns and needs. We sacrificed some amount of compatibility with the WSDOT study to accomplish these goals.

ECCA Roadside Litter Study

# Collection Protocol

*Procedures, Sampled Areas & Field Stages*

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Highway 14 Corridor · Clark & Skamania Counties, WA

# Why This Protocol Exists

**Ensure data consistency** across all sampled locations and volunteer teams, regardless of who collects on a given day.

**Minimize risk and complexity** during roadside collection by keeping categorization off-site.

**Capture litter composition data** needed to answer ECCA's core research questions on plastic pollution, alcohol-related litter, unsecured-load debris, and accumulation rates.

**Maintain methodological compatibility** with the 2022 Washington Statewide Litter Study to enable meaningful regional comparisons.

# 8 Segments · 5 Highway Strata · 3 Ramps

5  
Highway  
Strata

3  
Ramp  
Segments

~3  
Month  
Accumulation

## Accumulation Rule

Each segment must remain **uncleaned for ~3 months** before each collection event. This interval ensures statistically meaningful accumulation while not significantly delaying ECCA's ongoing cleanup commitments.

## Segment IDs

H1-8 Camas North

H2-1 Camas South

H3-3 Washougal Xtra (S)

H4-10 Washougal North

H5-5 Washougal South

*014Q101046 Vancouver S on-ramp*

*014S101194 Camas N on-ramp*

*014P101430 Washougal S off-ramp*

# The Protocol in Three Stages

1

## Preparation

*Before any roadside work begins*

- Confirm ~3-month accumulation period has elapsed
- Assemble and check all materials
- Brief volunteers on boundaries & safety

2

## Roadside Collection

*Speed and safety; no sorting on-site*

- Collect all litter within segment boundaries only
- Label every bag with Segment ID before filling
- Record large/bulky items on separate sheet

3

## Transport & Initial Weighing

*All bags → central sorting location*

- Record total segment weight before opening any bag
- Log: Segment ID, date, and combined weight
- Weight cross-checks item-level data in later sorting

# Before Leaving for the Field

## Materials Checklist

- Collection bags — labeled by Segment ID
- Large-item recording sheet
- Portable scale
- Applet device (charged & tested) for data entry
- Segment boundary info (map links from sampling results table)
- Hi-vis vests, gloves, litter picker tools

## Volunteer Briefing

### Exact segment boundaries

Confirm start/end points for today's collection using Google Maps links.

### Bag separation

Bags from different segments must be kept physically separate at all times — from collection through transport.

### Safety requirements

Hi-vis vests must be worn. Follow all roadside safety procedures before stepping outside the vehicle.

# Collect Fast, Stay Safe, Don't Sort

## Boundaries

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Collect only within designated start/end points confirmed by Google Maps links. Straying into adjacent areas changes the effective sample length and invalidates cross-segment comparisons.

## Bag Labeling

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Label every bag with the Segment ID before filling (e.g., "H1-8" or "014Q101046"). Never mix bags from different segments — keep them physically separate from collection through transport.

## Large / Bulky Items

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Items too large for standard bags (tires, lumber, construction debris) are documented on the large-item recording sheet: description, estimated size/weight, and likely source category.

***Key Rule: No item-by-item categorization takes place at the roadside — bag everything, label it, and categorize at the sorting location.***

# Weight First — Before Any Bag Is Opened

1

## Transport to Central Sorting Location

All bags from the segment travel together. Bags from different segments remain physically separate throughout transit.

2

## Weigh Before Opening

Record total combined weight of all segment bags before any bag is opened. This prevents loss or miscounting during sorting.

3

## Log Required Entries

Before sorting begins, the project data spreadsheet must include:

- Segment ID
- Collection date
- Total weight (kg or lb — consistent unit throughout the study)

# Key Reminders for Every Collection Event



## Wait the full ~3 months

Never collect a segment early — it breaks temporal comparisons.



## Stay within boundaries

Google Maps links confirm exact start/end points.



## Weigh before opening

Total segment weight is logged before any bag is opened.



## Label bags before filling

Segment ID on every bag before the first item goes in.



## No roadside sorting

All categorization happens at the central sorting location only.



## Log three fields

Segment ID · Collection date · Total weight (consistent units).

ECCA Roadside Litter Study

# Categorization Protocol

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# Why This Protocol Exists

**Provide consistent data collection** across all sampled segments and volunteer teams

**Record each litter item individually** using the app to improve accuracy and organization.

**Identify litter types and sources** such as plastic waste, beverage containers, and unsecured-load debris.

**Support future analysis and comparisons** by using a standardized categorization system.

## The Categorization Process in Three Stages

1

### Initial Weighing

*Before categorization begins*

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- Record total litter weight
- Record collection date and segment ID
- Record total plastic weight if possible

2

### Item-by-Item Categorization

*Each item is entered into the app*

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- Examine one item at a time
- Record item information in the app
- Classify by material and function
- Continue until all items are recorded

3

### Category Weighing

*After categorization is complete*

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- Weigh selected categories if needed
- Record total plastic weight
- Document large or unusual debris separately

## STAGE 1: Before Categorization Begins

### Information to Record

- Segment ID
- Collection Date
- Total Litter Weight
- Total Plastic Weight (if available)

### Volunteer Briefing

#### Why Record Weight First?

##### Complete Record

Record the total weight before any sorting or categorization begins.

##### Consistent Measurement

Provides a baseline measurement for each sampled segment.

##### Better Data Quality

Helps ensure all collected litter is accounted for during the categorization process.

## Stage 2: Item-by-Item Categorization

### Examine the Item

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Volunteers examine one litter item at a time and identify its main characteristics.

### Record Information in the App

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Information about the item is entered into the app, including its material, function, and other details.

### Assign Categories

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Each item is classified using:

- Material Category
- Functional Category
- Likely Source
- Brand Information (if available)
- Size Information (optional)

**Example:** A coffee cup may be classified as **Paper**, **Beverage Container**, and **Intentional Litter**

## STAGE 3: Weigh Selected Categories

1

### After Categorization

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All items have already been recorded and classified in the app before any additional weighing takes place.

2

### Record Key Measurements

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Current measurements include:

- Total Litter Weight
- Total Plastic Weight

3

### Document Large Debris

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Large or unusual items that do not fit easily into collection bags are recorded separately with additional notes.

**Current focus:** The main measurements of interest are total litter weight and total plastic weight.

# Conclusions & Next Steps

## What This Study Delivers

### A replicable sampling design

8 segments (5 highway + 3 ramps) chosen via stratified random sampling, with defined weights for total litter estimation.

### A consistent collection protocol

Three-stage field procedure — preparation, roadside collection, and initial weighing — standardized across all volunteer teams.

### An item-level categorization system

Applet-based data entry covering material, function, likely source, brand, and size/fragmentation for every item.

### Statewide comparability

Methodology aligned with the 2022 WA Statewide Litter Study, enabling future regional benchmarking.

## Recommended Next Steps

1

### Conduct the first collection round

Each sampled segment must be left uncleaned for ~3 months before collection begins.

2

### Sort and categorize collected items

Use the applet to record all item characteristics at the central sorting location.

3

### Investigate strata consolidation

With only one sample per highway stratum, combining strata where data supports it will improve analytical power.

4

### Plan for multi-round comparison

Repeated collections across time are the path to understanding litter accumulation rates.